



Site Map · Level 2 build

Trench both ways from the panel: a second trench, drawn point by point, each one dug once. ClearSky-OMEGA

Trench both ways from the panel

A Level 2 panel on a wall often feeds stalls to its left *and* its right, or a second row behind the first. The build now draws a second trench off the first one, point by point, so the two go their own way from the panel. Each trench is dug once and nothing overlaps; every charger's conduit follows its own trench back to the panel, and the trench total counts each excavation once.

This guide runs one job from the Build tab to the finished drawing. Keys: **Enter** finishes a trench, **Backspace** removes its last point, **T** starts another trench, **Esc** backs out of that second trench. A click on any conduit shows the trench it rides, with points to drag.

1 Start the Level 2 build

Build tab > **Level 2**. Choose the charger model, how many, and where the service starts, then **Start Placing**. This job has four chargers.

Level 2 EV Charging — Guided Build

CHARGER MODEL

Autel AC Pro — 19.2kW / 80A / 1 port

Nameplate: 48A nameplate

How many: 4

Spare conduit: 0

☒ Place bollards at each charger

LOAD CHECK

4 x Autel AC Pro @ 48A - 60A circuit each - 46.1 kW connected
Panel and existing service must both carry 240A of new branch load — verify before this reaches an estimate.

START FROM

Existing meter — tap the service that is there

You'll place: ● Existing Meter Bank → draw the trench → ● Outdoor Panel → ● EVSE
Conduit auto-routes: interior feed in yellow, trenched branch in cyan. Set the trench to asphalt after it draws if it crosses pavement.

Start Placing

The Level 2 dialog. Four chargers, an existing meter bank as the start.



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2 Place the meter, draw the feed, place the panel

Click the map to drop the meter bank. Draw the feed run to where the outdoor panel goes (click points, **Enter**), then click on that run to place the panel. The guide panel on the right tracks each step.

The meter, the interior feed (yellow) and the outdoor panel on its end. Next: the branch trench.



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3 Draw the first trench and drop the chargers on it

Draw the branch trench from the panel along the first row of stalls, press **Enter**, then click on the trench to drop each charger. A click past the end extends the trench to that stall.

The screenshot shows the ClearSky-OMEGA software interface for the Level 2 build. The main workspace displays a site map with a trench layout. A legend on the right lists equipment and conduit types. A guide panel on the right provides instructions for drawing the trench and dropping chargers.

Legend:

- EQUIPMENT**
 - Existing Meter Bank: 240/120V 1Ø 3-wire — existing, verify available capacity
 - Outdoor Panel / Disconnect: 60A 240V 1Ø - NEMA 3R - 4 x EVSE branch
 - Autel AC Pro x2: 19 kW - 48A - 1 port
 - Bollard
- CONDUIT**
 - Interior conduit (240V 1Ø): 1 run - 35.1 ft
 - Trench (panel - EVSE): 2 runs - 100.2 ft of conduit - in trench
 - Trench 70.0 ft - dug once

Guide Panel:

- Existing Meter Bank
- Outdoor Panel / Disconnect
- Chargers (2 / 4)
- Click anywhere on the BRANCH trench to drop a charger — it snaps to the line. Click past its end and the trench extends to that stall; click off it and a spur runs there from the nearest point. Going the other way, or around something? Draw another trench (T) from the panel or anywhere on the run. 2 left. Press Enter to finish.
- ≠ Flip (F)
- ↻ Rotate (R)
- + Draw another trench (T)
- ✓ Finish Build (Enter)
- Cancel (Esc)

Bottom Bar:

Click on the branch trench to drop each charger (2 left) — click past its end to extend it, T draws another ...

The first trench heads east from the panel with two chargers on it. Two left to place.

4 Press T for another trench

Press **T**, or click **+ Draw another trench** in the guide panel. Move the mouse toward the panel: a ring shows where the new trench will begin. The first click always lands on the trench already drawn — at the panel, or anywhere along it — never beside it.



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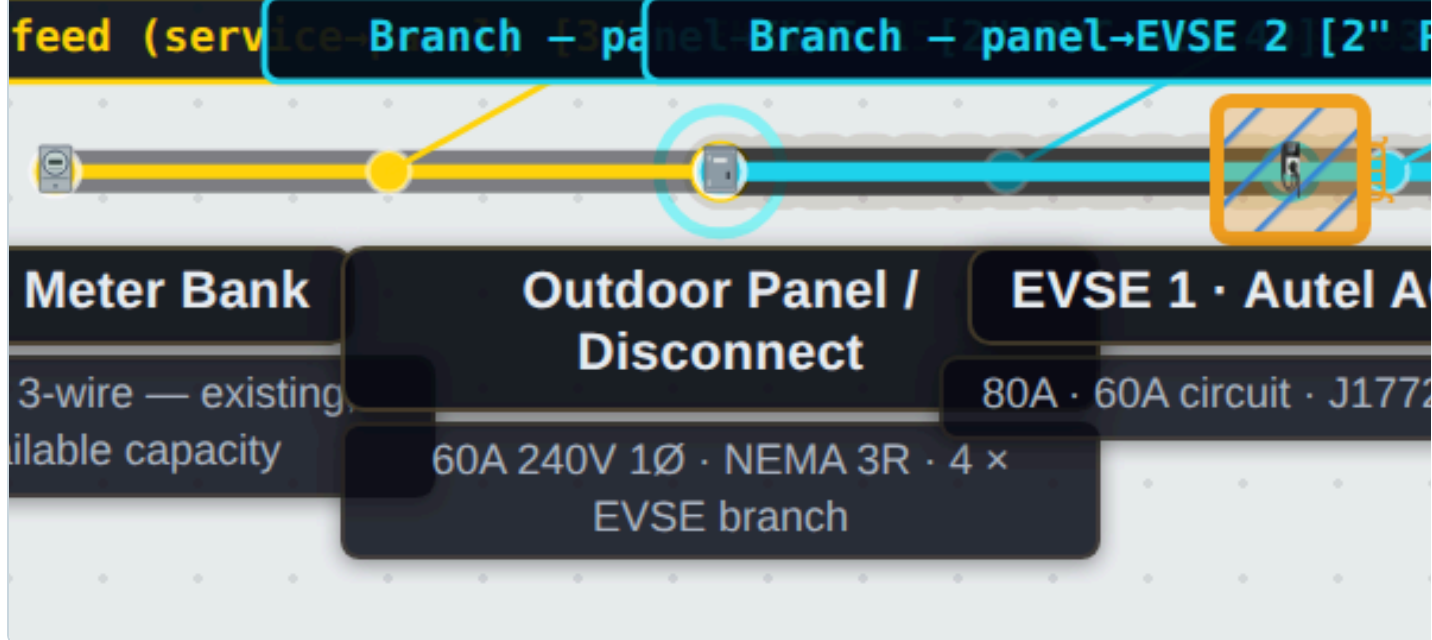
The screenshot shows the 'Level 2 Build' interface in ClearSky-OMEGA. The top menu bar includes File, Edit, Build, Compute, Insert, Draw, Annotate, Analyze, Estimate, View, Output, Settings, and The Ladder. The 'Build' tab is active, showing various tool icons for Site Setup, Level 2, DER Build, Auto Layout, BESS Build, Full Topology, DCFC Build, Building Designer, Design with AI, Engineering Build, and Solar + Storage. The main workspace displays a site map with a trench layout. A tooltip explains the second trench: 'Draw another BRANCH trench: the first click lands on the trench you drew, then points to route it. Enter fin...'. A right-hand panel shows a list of components: Existing Meter Bank, Outdoor Panel / Disconnect, and Chargers (2 / 4). A legend at the bottom right defines symbols for equipment (Existing Meter Bank, Outdoor Panel / Disconnect, Autel AC Pro x2, Bollard) and conduit (Interior conduit (240V 1Ø), Trench (panel - EVSE), Trench 70.0 ft - dug once).

After T the guide panel explains the second trench; the cursor is over the panel.



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Close up: the ring on the panel is where the second trench will begin.



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5 Route it, with bends, and press Enter

Click the panel, then each bend along the way. **Backspace** takes the last point back; **Esc** or **Back to chargers** drops the whole draft. **Enter** files the trench.

The screenshot shows the ClearSky-OMEGA software interface for the 'Level 2 build' phase. The main workspace displays a site map with a trench layout. A legend on the right lists equipment and conduit types. A pop-up window on the right provides instructions for drawing a branch trench.

LEGEND

EQUIPMENT

- Existing Meter Bank
240/120V 1Ø 3-wire — existing, verify available
- Outdoor Panel / Disconnect
60A 240V 1Ø · NEMA 3R · 4 × EVSE branch
- Autel AC Pro x2
19 kW · 48A · 1 port
- Bollard

CONDUIT

- Interior conduit (240V 1Ø)
1 run · 35.1 ft
- Trench (panel - EVSE)
2 runs · 100.2 ft of conduit · in trench
- Trench 70.0 ft · dug once

Draw another BRANCH trench: the first click lands on the trench you drew, then points to route it. Enter fin...

Draw another BRANCH trench: your first click lands on the trench already drawn — at the panel, or anywhere along it — then click points to route it (3 pts) and press Enter. Backspace removes the last point; Esc goes back to the chargers.

Buttons:

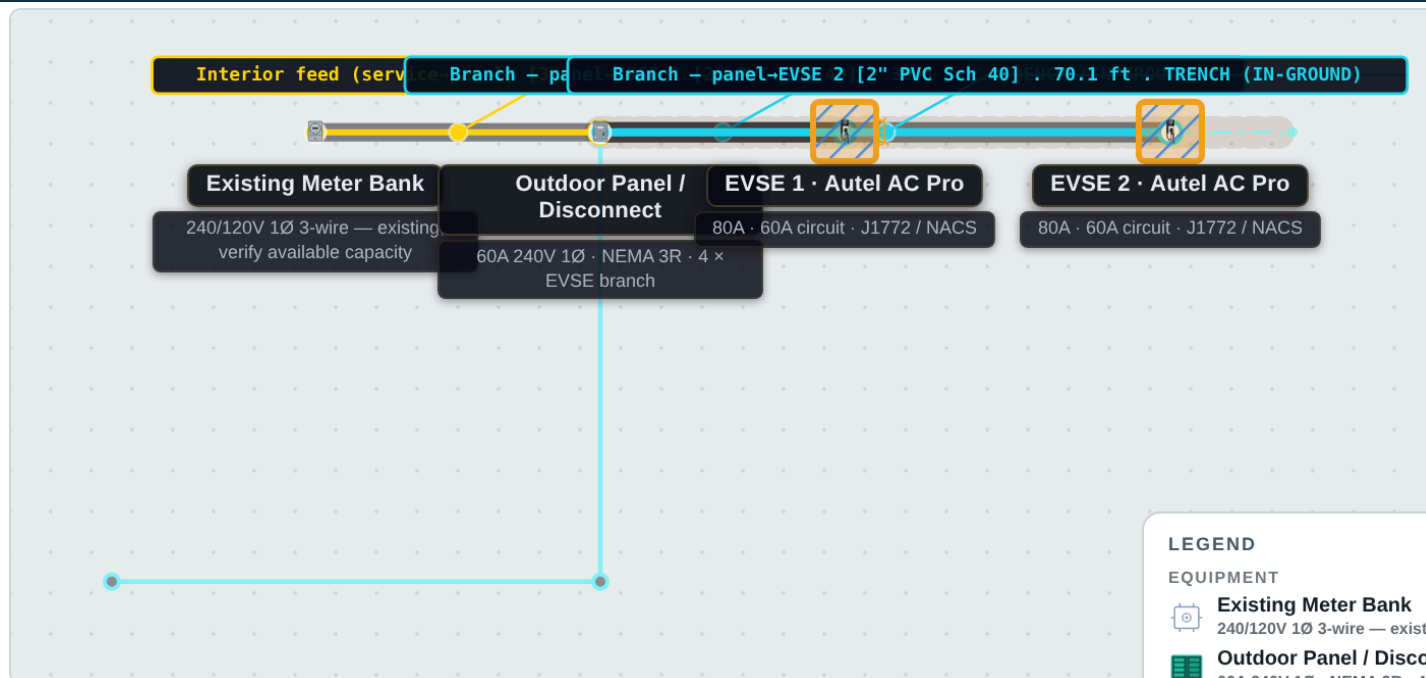
- ✓ Finish Run (Enter)
- ↶ Undo Point (Backspace)
- Surface: in-ground
- ← Back to chargers (Esc)
- Cancel (Esc)

The second trench drawn south from the panel, then west: three points, still a draft (the guide panel counts them).



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Close up: the draft leaves the panel, turns, and runs along the second row. Enter files it.

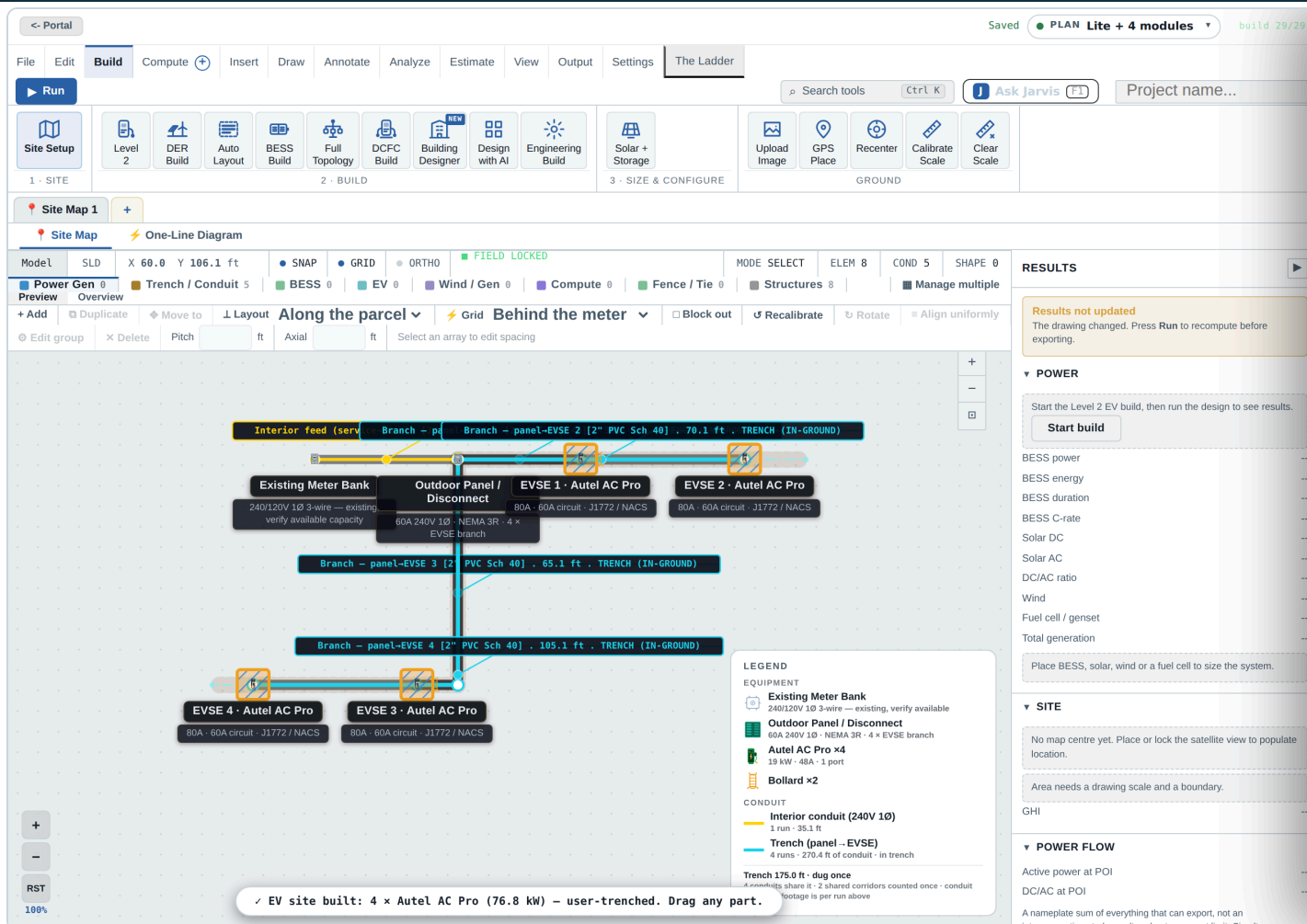
6 Drop the rest of the chargers on the new trench

Click on either trench to place the remaining chargers. Each one's conduit follows the trench it sits on back to the panel. The build finishes on its own when the last charger is down.



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Trench both ways from the panel: a second trench, drawn point by point, each one dug once. ClearSky-OMEGA



Four chargers on two trenches leaving the same panel. Nothing overlaps, nothing is dug twice.

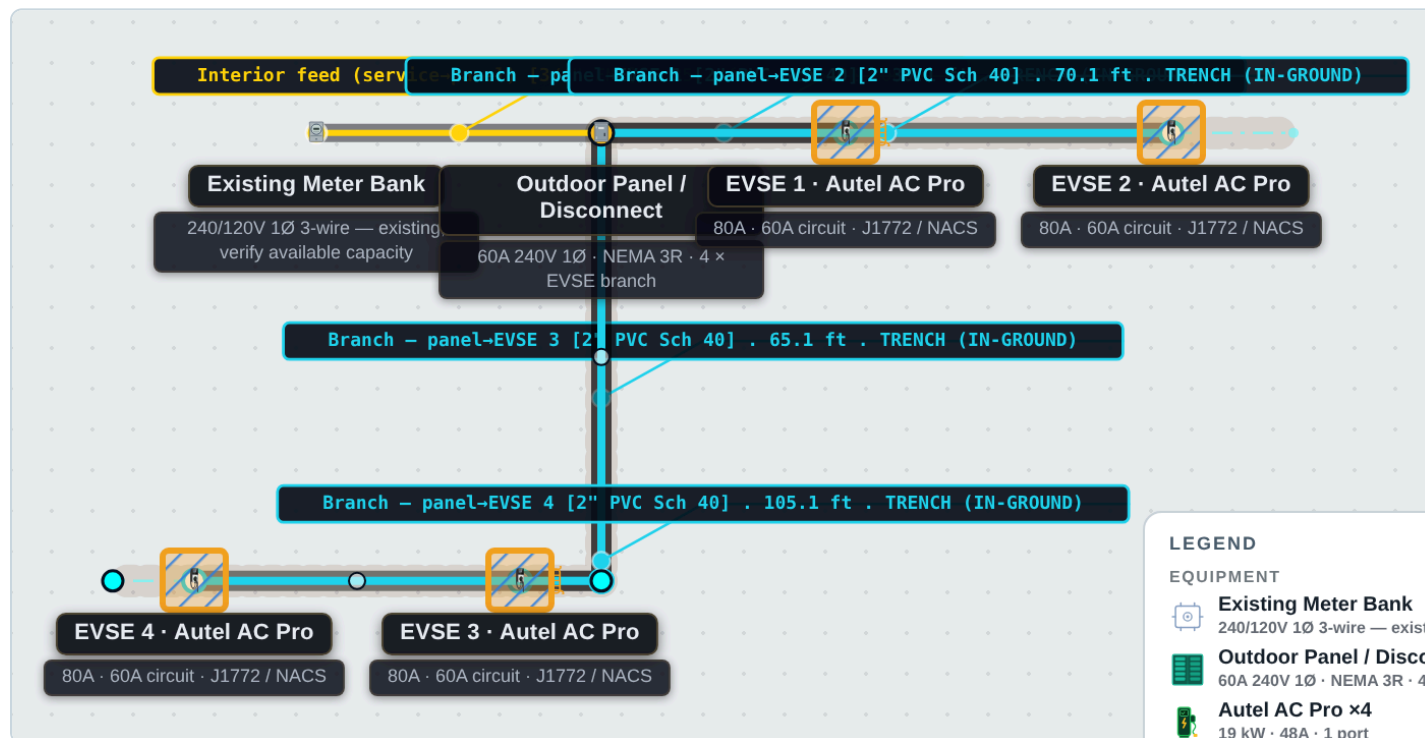


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7 Move a trench after it is drawn

Clicked the chargers first and the trench went the wrong way? Click any conduit on that trench: the trench's own points appear. Drag a point, or drag a faint midpoint to add a bend, and every conduit laid in that trench follows. Double-click a point to remove it, **Esc** to finish, undo takes it back.

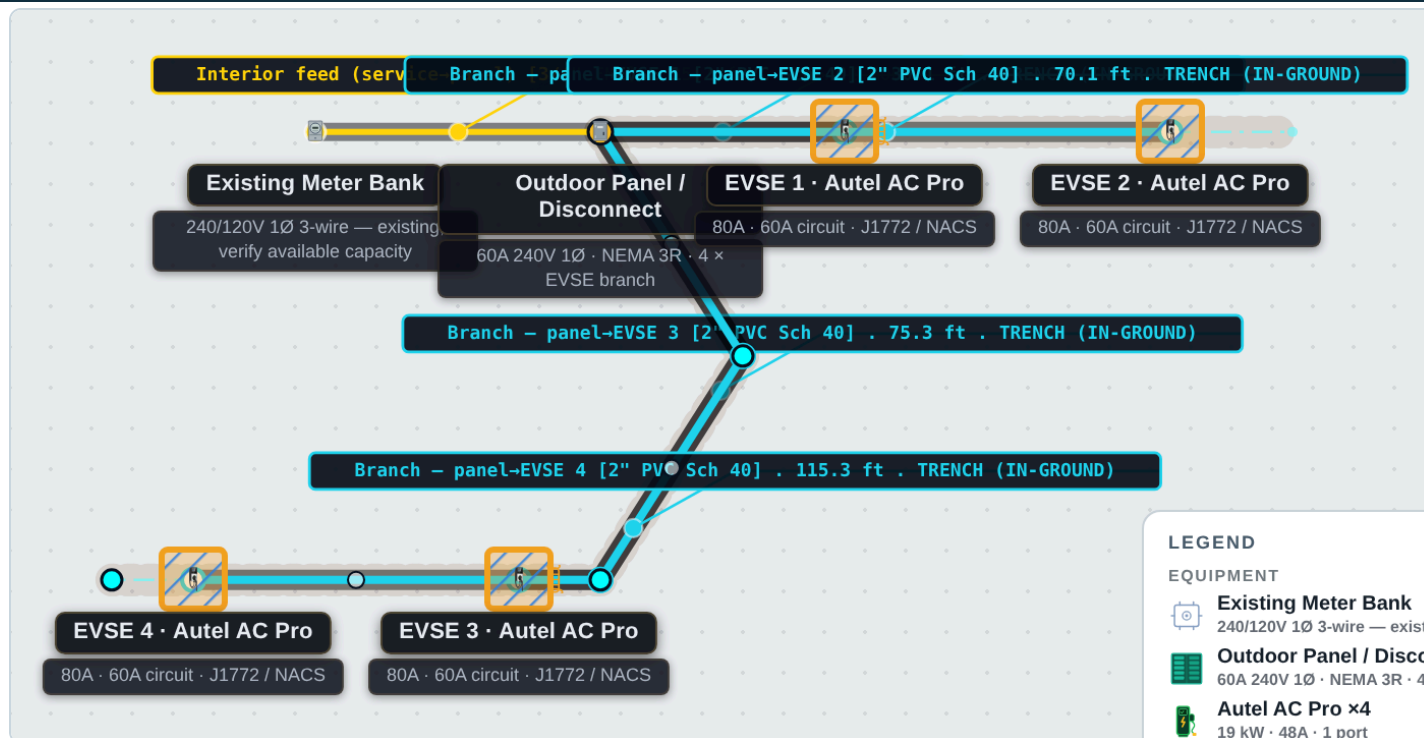


A click on a conduit shows the trench it rides: its points, and a faint midpoint on each segment.



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The midpoint dragged into a bend. Both chargers' conduits moved with the trench: one line, no ghost. Open a trench whose conduits were dragged apart one by one and they are laid back on it.



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8 What the totals say

The legend counts the trench as excavation: each drawn trench once, the first trench only as far as the second one taps it, and the conduit inside them separately. The conduit schedule and the BOM read the same numbers.

LEGEND

EQUIPMENT



Existing Meter Bank

240/120V 1Ø 3-wire — existing, verify available



Outdoor Panel / Disconnect

60A 240V 1Ø · NEMA 3R · 4 × EVSE branch



Autel AC Pro ×4

19 kW · 48A · 1 port



Bollard ×2

CONDUIT



Interior conduit (240V 1Ø)

1 run · 35.1 ft



Trench (panel → EVSE)

4 runs · 290.8 ft of conduit · in trench

Trench 185.2 ft · dug once

4 conduits share it · 2 shared corridors counted once · conduit and cable footage is per run above

The legend after the build: the branch conduit runs, and the trench dug once.

Good to know



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- A charger clicked *off* the trench still gets a straight spur from the nearest point. Use **T** when the route has to bend before the first stall, or should leave from a particular point.
- The second trench can start anywhere on the first one, not only at the panel: a tee off the middle of a row works the same way.
- The surface button (in-ground or sawcut concrete) applies to the trench being drawn.
- **Esc** while drawing the *second* trench goes back to the chargers; while drawing the first trench it cancels the build, keeping what was placed.
- A charger clicked across the building makes a straight spur. Click that conduit and drag the trench's points round the building; both chargers' conduits follow, and there is only ever one trench for that side.
- The same controls are in the DCFC build.